

AfriVTON-Bench: Evaluating the Robustness of Virtual Try-On Models to African Textile Patterns and Garment Structures

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Abstract

Virtual Try-On (VTON) systems have advanced considerably in recent years, yet their generalisation across global fashion diversity remains constrained by training corpora that skew heavily toward Western garment aesthetics. We introduce AfriVTON-Bench, a benchmark designed to evaluate VTON robustness on African garments, a domain characterised by high-frequency repeating wax-print and woven patterns as well as voluminous non-Western draping silhouettes that lie far outside the distribution of existing evaluation suites. We curate 111 African garment images spanning 16 categories across 7 countries and 5 sub-regions, pair them with a 50-image Western control set drawn from VITON-HD, and evaluate two state-of-the-art architectures across 1,012 person-garment pairs. Both models exhibit pronounced garment-person reconstruction asymmetry: FASHN-VTON preserves person identity $3.63\times$ more faithfully than garment texture (pLPIPS=0.179 vs. gLPIPS=0.650), while LEFFA shows a gap of $4.27\times$ (pLPIPS=0.156 vs. gLPIPS=0.664). African garments suffer a structural fidelity deficit of 9–10 percentage points in gSSIM relative to Western controls, with Kente (gSSIM=0.460) and Ankara (gSSIM=0.474) degrading most severely. We identify three systematic failure modes – pattern dissolution, drape collapse, and tonal drift – that cut across both architectures, indicating that the root cause lies in the shared Western training distribution rather than any particular design choice.

1 Introduction

Virtual Try-On (VTON) systems, which digitally composite a target garment onto a reference person image, have undergone rapid capability gains driven by diffusion-based architectures. Systems such as OOTDiffusion [Xu *et al.*, 2024], IDM-VTON [Choi *et al.*, 2024], CatVTON [Chong *et al.*, 2025], and FitDiT [Jiang *et al.*, 2024] now produce photorealistic synthesis results on the VITON-HD benchmark [Choi *et al.*, 2021], with perceptual metrics approaching those of

ground-truth photograph pairs. These capabilities have been deployed in online fashion retail, where try-on tools reduce return rates by enabling customers to visualise fit without physical garments.

The apparent quality gains, however, are tightly coupled to a narrow training distribution. VITON-HD [Choi *et al.*, 2021] and DressCode [Morelli *et al.*, 2023b], the two benchmarks on which virtually all contemporary models are trained and evaluated, were curated from Western fashion e-commerce platforms. The garments in these datasets are overwhelmingly Western in silhouette and lightly textured in fabric, and the associated subjects cluster in the lighter half of perceptual skin tone scales. Published performance figures may therefore substantially overestimate system quality for users and garment types outside this distribution. This is a well-known instance of the dataset bias problem formalised by Buolamwini and Gebru [2018], who showed that aggregate benchmark performance routinely conceals systematic degradation across underrepresented subgroups.

African textile garments pose a structurally distinct challenge for existing VTON pipelines. Fabrics such as Ankara, Kente, Adire, and Aso-Oke carry high-frequency repeating motifs with strong geometric regularity and high local contrast – properties that expose known failure modes in both spatial warping and diffusion synthesis stages [Jiang *et al.*, 2024]. Traditional garments including Agbada, Boubou, and Dashiki have voluminous, non-fitted silhouettes that differ substantially from the body-congruent clothing dominant in existing benchmarks. The subjects most naturally associated with these garments also tend to have darker skin tones, a demographic that the fairness literature has consistently found to bear disproportionate performance penalties in generative vision systems [Buolamwini and Gebru, 2018].

Work on African fashion in computer vision remains sparse. AFRIFASHION1600 [Oyewusi *et al.*, 2021], the only dedicated African fashion dataset for computer vision, targets classification rather than generation and lacks the garment-person pairs required for VTON evaluation. AFRIGAN [Salami *et al.*, 2021] applied generative modelling to Ankara fabric synthesis but addresses an unconditional texture task that is structurally distinct from person-conditioned compositing. Omotayo *et al.* [2024] report that fewer than 0.06% of computer vision papers address African-specific visual domains, a figure that reflects the near-absence of African insti-

tutions from benchmark construction in the field. No existing VTON work examines model behaviour on African textiles, garment structures, or the skin tone distribution of African subjects.

This paper addresses that gap through four contributions: a systematic audit of garment pattern complexity and skin tone distributions within VITON-HD and DressCode; AfriVTON-Bench, a curated benchmark spanning 16 African garment categories paired with subjects covering the full Monk Skin Tone scale [Schumann *et al.*, 2023]; a stratified robustness evaluation of FASHN-VTON and LEFFA using gLPIPS and pLPIPS [Zhang *et al.*, 2018], gSSIM [Wang *et al.*, 2004], and DISTS_g [Ding *et al.*, 2022]; and tonal drift, a metric that quantifies the degree to which a VTON model alters the apparent skin tone of the subject during synthesis.

2 Related Work

2.1 Virtual Try-On: Architectures and Benchmarks

The virtual try-on problem requires rendering a target garment onto a reference person image while preserving identity, pose, and body geometry. Early neural approaches decomposed the task into a spatial warping stage followed by a blending stage, using thin-plate spline or appearance flow transformations to align the garment to the target pose [Han *et al.*, 2018; Wang *et al.*, 2018]. These GAN-based pipelines exhibited characteristic failure modes under large pose variation, attributable to the limited expressivity of globally smooth deformation fields [Lee *et al.*, 2022].

The shift to diffusion-based architectures substantially raised synthesis quality. VITON-HD [Choi *et al.*, 2021] introduced a misalignment-aware normalisation mechanism conditioned on semantic garment-body alignment, establishing the benchmark that subsequent work builds upon. LaDI-VTON [Morelli *et al.*, 2023a] replaced the GAN generator with a Latent Diffusion Model conditioned on garment features through CLIP-based textual inversion, bridging the warping and fully generative paradigms. OOTDiffusion [Xu *et al.*, 2024] enables garment-detail preservation without explicit warping by entangling garment and person representations through a parallel garment encoder. IDM-VTON [Choi *et al.*, 2024] introduces IP-Adapter conditioning on a Diffusion Transformer backbone, decoupling garment appearance from pose encoding. CatVTON [Chong *et al.*, 2025] demonstrates that concatenating garment and person images spatially before the diffusion pass achieves competitive quality while substantially reducing architectural complexity. FitDiT [Jiang *et al.*, 2024] extends the DiT paradigm with a frequency-domain texture loss that penalises high-frequency degradation in the Fourier domain, motivated by the inadequacy of standard perceptual losses for preserving fine-grained repeating patterns.

The VTON benchmark landscape is narrow. VITON-HD [Choi *et al.*, 2021] comprises roughly 13,000 upper-body garment-person pairs from Western e-commerce platforms. DressCode [Morelli *et al.*, 2023b] extends coverage to approximately 53,000 full-body and lower-body pairs across three garment categories but shares the same Western prove-

nance. Neither dataset has been audited for pattern complexity distribution or subject skin tone composition.

2.2 Bias and Fairness in Generative Vision

The systematic auditing of vision systems for demographic bias was formalised by Buolamwini and Gebru [2018], whose Gender Shades study demonstrated differential error rates across gender and skin tone groups in commercial face analysis APIs. Their methodological contribution – stratified evaluation across independently annotated demographic subgroups to disaggregate aggregate metrics – is the audit framework this work inherits and applies to the VTON setting.

The Monk Skin Tone (MST) scale used in our study was developed by Schumann *et al.* [2023] as a perceptually uniform alternative to the Fitzpatrick scale [Fitzpatrick, 1988], which was designed for clinical dermatology and compresses perceptual variation at darker tones. Schumann *et al.* [2023] demonstrate substantially higher inter-annotator agreement across diverse annotator pools, providing a principled basis for the skin tone stratification in our evaluation.

Bias in generative models presents distinct challenges relative to discriminative systems. Rather than manifesting as differential accuracy, it may appear as differential synthesis quality, reduced fidelity to input attributes, or systematic hallucination of attributes absent from the conditioning input [Cho *et al.*, 2023]. In the VTON context, a model may alter the apparent skin tone of the subject during synthesis in ways that are invisible to standard perceptual metrics. We term this tonal drift. DS-VTON [Sun and others, 2025] is the closest related work, introducing a skin segmentation-aware refinement loss that implicitly acknowledges skin fidelity as an active concern, but it does not provide skin-tone-stratified results or characterise tonal drift as a distinct failure mode.

2.3 African Fashion in Computer Vision

Computer vision research on African fashion remains at an early stage. AFRIFASHION1600 [Oyewusi *et al.*, 2021] was designed for supervised classification across fourteen garment categories; its scale and design make it unsuitable for generative VTON evaluation, which requires high-resolution garment-person pairs with controlled pose and lighting. AFRIGAN [Salami *et al.*, 2021] is the only prior application of deep generative modelling to African fashion, proposing a GAN-based architecture for Ankara pattern synthesis. That work demonstrates the visual distinctiveness of African textiles as a generative challenge, but its unconditional texture synthesis task is structurally different from person-conditioned garment compositing.

The structural deficit underlying these gaps is quantified by Omotayo *et al.* [2024], whose survey finds that fewer than 0.06% of computer vision publications address African-specific visual domains. The near-absence of African institutions from programme committees at CVPR, NeurIPS, ICCV, and ECCV is identified as a primary mechanism through which African visual culture is excluded from benchmark construction and task definition. For VTON, the consequence is that the garment types and skin tones most relevant to a large share of the world’s population are entirely absent from the benchmarks used to measure model progress.

2.4 Texture and Pattern Preservation

Preserving high-frequency texture under spatial transformation is a longstanding challenge in generative synthesis. Johnson et al. [2016] established that minimising feature-space distance from a pretrained VGG network substantially improves texture fidelity over pixel-wise ℓ^2 losses, which weight all spatial frequencies equally and produce characteristic blurring. Zhang et al. [2018] formalised this as the LPIPS metric, providing a perceptual distance measure correlated with human quality judgements across a wide range of distortion types. FID [Heusel et al., 2017] captures complementary distributional properties by measuring the distance between synthesised and real image populations in Inception feature space.

In VTON pipelines, warping-based components are the primary locus of high-frequency texture loss. Optical-flow and TPS modules estimate globally smooth deformation fields that apply the same transformation across all spatial frequencies; high-frequency motifs requiring locally precise alignment are therefore systematically blurred [Han et al., 2018; Wang et al., 2018]. Diffusion-based systems partially address this by replacing explicit warping with learned synthesis, but remain bounded by the frequency content of their training data. FitDiT [Jiang et al., 2024] targets this directly through Fourier-domain supervision, though whether that supervision transfers to the extreme pattern densities of Ankara and Kente textiles – absent from its training data – is an open empirical question this benchmark addresses.

3 Methodology

3.1 Benchmark Construction

AfriVTON-Bench is a curated evaluation dataset designed to expose distribution-shift failures in VTON systems trained on Western fashion corpora. Construction proceeded in two stages: garment collection and person image collection.

We assembled 111 garment images spanning 16 categories across 7 African countries and 5 geographic sub-regions: Nigeria (Ankara, Agbada, Aso-Oke, Lace), Ghana (Kente, Batakari), Senegal (Boubou, Kaftan), Ethiopia (Habesha Kemis, Netela), Kenya (Kikoi, Shuka), South Africa (Shweshwe), and Morocco (Kaftan, Djellaba). Images were sourced through automated web scraping via DuckDuckGo, Wikimedia Commons, and Bing using the iCrawler library, supplemented by targeted manual curation for categories where automated retrieval yielded insufficient quality, particularly Nigerian Agbada and South African beadwork. A Western control set of 50 garment images was drawn from VITON-HD [Choi et al., 2021] to serve as an in-distribution baseline. Each garment was annotated with country of origin, geographic region, garment type, visual complexity tier, pattern type, and drape type. Gender suitability labels were assigned through Gemini-based classification and verified manually.

Fifteen person model images were sourced targeting diverse African subjects. Each was annotated with Monk Skin Tone Scale values [Schumann et al., 2023], predominantly Monk 7, and gender labels to enable future stratified analysis across skin tone groups.

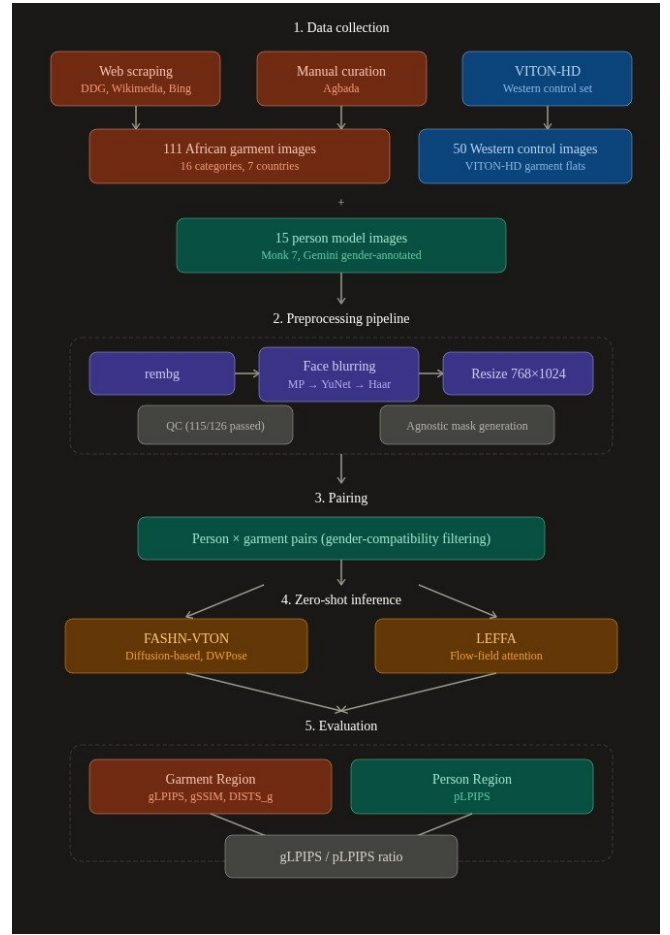


Figure 1: Overview of the AfriVTON-Bench pipeline. Garments are collected through web scraping and manual curation and paired with person model images through gender-compatibility filtering, before zero-shot inference with FASHN-VTON and LEFFA.

3.2 Preprocessing Pipeline

All images underwent a standardised preprocessing pipeline before inference. Garment images were processed using rembg for automatic background removal and padded to 768×1024 with white backgrounds. A quality control step checked corner-pixel whiteness; one Ethiopian Netela image was flagged for insufficient coverage. Western control garments bypassed background removal but were resized to the same resolution.

To protect personally identifiable information in scraped images, a cascading multi-detector face blurring pipeline was applied. Three detectors are employed in priority order: MediaPipe Face Detection (minimum confidence 0.35, chosen to reduce false negatives on darker skin tones), YuNet, and Haar Cascade as a final fallback. A two-pass strategy applies detection on the full image and on a 2× upscaled top-half crop, the latter to catch small faces common in full-body fashion photography. Where no face is detected by any method, Gaussian blur is applied to the top 15% of the image as a safe-zone fallback.

Of 126 images evaluated by the automated QC pass, 115

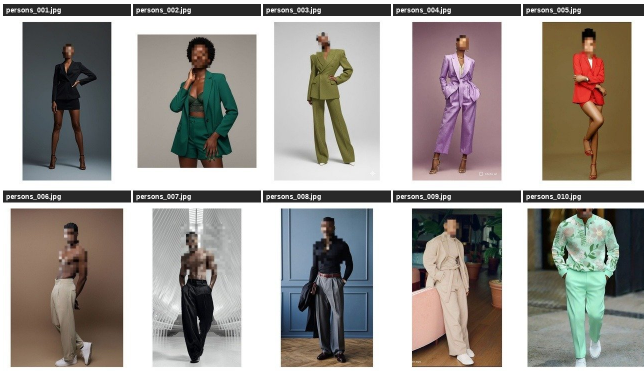


Figure 2: Representative subset of the 15 person model images. Faces blurred for privacy. Persons 001–005 are female-presenting; persons 006–015 are male-presenting. All images are standardised to 768×1024 resolution.

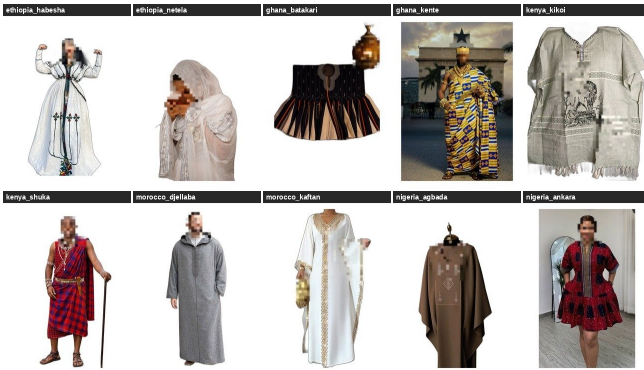


Figure 3: Representative samples from the 16 African garment categories (left to right, top to bottom): Ethiopia Habesha Kemis, Ethiopia Netela, Ghana Batakari, Ghana Kente, Kenya Kikoi, Kenya Shuka, Morocco Djellaba, Morocco Kaftan, Nigeria Agbada, Nigeria Ankara. All images preprocessed to 768×1024 with white backgrounds; faces blurred where present.

passed. All 11 failures were Western control garments with insufficient visible garment area; no African garment image failed the check.

3.3 Model Selection and Inference

Two VTON architectures were selected to represent distinct design paradigms. FASHN-VTON is a diffusion-based model built on Stable Diffusion, employing DWPose for pose estimation and a segmentation-free inference mode. The `fashn-vton-1.5` checkpoint was used with 50 denoising timesteps, guidance scale 2.5, and seed 42 for reproducibility. LEFFA (Learning Flow Fields in Attention) learns spatial warping within the attention mechanism of a diffusion model; checkpoints were obtained from the official HuggingFace repository. All experiments ran on an NVIDIA Tesla T4 GPU via Google Colab.

For each person image, garments were sampled with gender-compatibility filtering. FASHN-VTON processed all valid pairs, yielding 792 outputs; LEFFA used a reduced sampling strategy due to higher per-sample inference cost, yield-

Table 1: Aggregate evaluation metrics across all 1,012 person-garment pairs. The g/p Ratio column (gLPIPS / pLPIPS) quantifies reconstruction asymmetry between the garment and person regions; a higher value indicates proportionally greater failure on garment texture relative to person identity preservation.

Model	gLPIPS ↓	gSSIM ↑	DISTS _g ↓	pLPIPS ↓	g/p Ratio
FASHN-VTON	0.650	0.586	0.312	0.179	3.63×
LEFFA	0.664	0.577	0.312	0.156	4.27×

ing 220 outputs. The full evaluation set comprises 1,012 try-on images across both models.

3.4 Evaluation Metrics

Four perceptual metrics were computed at image level on 768×1024 images using PyTorch with CUDA acceleration. Garment LPIPS (gLPIPS) is the Learned Perceptual Image Patch Similarity [Zhang *et al.*, 2018] between the try-on output and the reference garment, computed with an AlexNet backbone; lower values indicate better garment texture preservation. Garment SSIM (gSSIM) is the Structural Similarity Index [Wang *et al.*, 2004] between the try-on result and the garment image; it captures luminance, contrast, and structural correspondence, with higher values indicating better preservation. DISTS_g [Ding *et al.*, 2022] is computed on $[0, 1]$ -normalised tensors and is more sensitive to texture complexity than SSIM, providing a complementary view of fine-grained garment fidelity. Person LPIPS (pLPIPS) measures perceptual similarity between the try-on output and the original person image, quantifying preservation of body, pose, and non-garment appearance.

The gLPIPS-to-pLPIPS ratio is used throughout as a model-agnostic index of garment-specific failure. A high ratio means the model reconstructs person identity far more faithfully than garment texture, which is evidence that the failure is concentrated in the garment region rather than in general synthesis quality.

4 Results and Discussion

4.1 Aggregate Performance

Table 1 presents results across all 1,012 pairs. The most consequential finding is the pronounced asymmetry between garment and person reconstruction. FASHN-VTON preserves person identity $3.63\times$ more faithfully than garment texture; LEFFA’s gap is wider at $4.27\times$. This is not primarily a synthesis quality problem in the conventional sense: both models render the person’s body and pose at reasonable perceptual fidelity, but fail specifically in the garment region. Despite representing fundamentally different design paradigms – FASHN-VTON uses DWPose-conditioned diffusion while LEFFA learns warping within the attention mechanism – both architectures converge on near-identical garment distortion (gLPIPS 0.650 vs. 0.664; DISTS_g 0.312 vs. 0.312), pointing toward the shared training distribution as the root cause.

4.2 African vs. Western Garments

African garments score 9.4–9.6 percentage points lower in gSSIM than Western controls across both models (Table 2),

Table 2: Performance stratified by garment origin. The gSSIM gap between African and Western garments is 9.4–9.6 percentage points across both models, while gLPIPS values are broadly similar – indicating that structural integrity is disproportionately compromised on African textiles beyond what aggregate perceptual distance alone suggests.

Model	Origin	n	gLPIPS ↓	gSSIM ↑	pLPIPS ↓
FASHN-VTON	African	474	0.650	0.549	0.204
	Western	318	0.651	0.643	0.142
LEFFA	African	142	0.666	0.543	0.167
	Western	78	0.659	0.639	0.134

Table 3: Per-category results ranked by FASHN-VTON gSSIM (ascending). Western Control is the in-distribution reference baseline. † marks LEFFA cells with fewer than 3 samples.

Category	FASHN-VTON		LEFFA	
	gLPIPS	gSSIM	gLPIPS	gSSIM
Ghana Kente	0.654	0.460	0.674	0.437
Nigeria Ankara	0.711	0.474	0.710	0.477
Ghana Batakari	0.715	0.513	0.677	0.532
Kenya Kikoi	0.717	0.519	0.728	0.495
Kenya Shuka	0.657	0.542	0.729	0.504†
Nigeria Aso-Oke	0.664	0.542	0.629	0.574
Nigeria Lace	0.614	0.561	0.782	0.381†
Morocco Djellaba	0.618	0.575	0.632	0.545†
Nigeria Agbada	0.656	0.589	0.654	0.606
Ethiopia Netela	0.665	0.596	0.672	0.574
S.A. Shweshwe	0.586	0.621	0.601	0.625
Senegal Boubou	0.584	0.623	0.608	0.624
Senegal Kaftan	0.608	0.632	0.638	0.643
Morocco Kaftan	0.566	0.640	0.555	0.655
Ethiopia Habesha	0.572	0.658	0.612	0.646
Western Control	0.651	0.640	0.658	0.638

confirming that structural fidelity degrades substantially on African textiles. The gLPIPS values, by contrast, are broadly comparable across origins (approximately 0.650 for both). The dissociation between these two metrics is informative: overall perceptual patch distance is similarly poor regardless of garment type, yet the spatial organisation and geometric regularity of African garments are disproportionately destroyed during synthesis. DISTS_g is slightly higher for Western garments than for African ones (0.345 vs. 0.291 for FASHN-VTON), which reflects the metric’s greater sensitivity to texture complexity rather than geometric structure; higher-frequency African patterns produce lower raw DISTS scores even when structural coherence has already been lost.

4.3 Per-Category Analysis

Table 3 reveals a failure hierarchy that maps predictably onto visual complexity. Kente and Ankara occupy the bottom two positions in gSSIM for both models, representing a 17–19 percentage point deficit relative to the Western control. Both categories also carry the highest gLPIPS values, confirming co-occurring perceptual and structural degradation. The rank ordering is highly consistent across models (Spear-

man $\rho \approx 0.88$ for categories with adequate LEFFA samples), which strengthens the reading that the hierarchy reflects genuine garment-driven difficulty rather than sampling variation.

At the other end, Ethiopia Habesha Kemis, Morocco Kaftan, and Senegal Kaftan cluster at or above the Western control in gSSIM. This is a notable result: it confirms that the observed degradation is driven by garment visual complexity, not by African origin as such. Nigeria Agbada, a voluminous silhouette garment, performs better than its structural complexity might predict (FASHN-VTON: 0.589, LEFFA: 0.606), likely because the category images are predominantly loose over-robe styles rather than fully draped ceremonial variants, placing them closer to the structured-garment regime both models were trained on.

4.4 Systematic Failure Modes

Three failure modes are observed consistently across both architectures, grounded in both quantitative results and qualitative inspection of model outputs. Figure 4 shows representative try-on outputs for four garment categories alongside their input person and reference garment images.

Pattern dissolution is the most severe failure mode, affecting high-frequency repeating motifs in Ankara wax-print, Kente narrow-strip weave, and Aso-Oke hand-woven patterns. Both models smear or homogenise the motifs during warping, producing outputs in which the textile’s visual identity is substantially lost. The Kente row in Figure 4 illustrates the effect; LEFFA additionally collapses the draped silhouette into fitted trousers, combining pattern dissolution with drape collapse. Maintaining periodic regularity across a spatially transformed field requires considerably more precise flow estimation than is needed for smooth, solid-coloured fabrics, and neither model’s training distribution prepared it for that demand.

Drape collapse affects voluminous non-tailored garments such as Agbada and Boubou. These garments feature flowing multi-layered fabric that extends beyond the body contour, and both models collapse such silhouettes inward into narrower, body-congruent outlines. The Agbada row in Figure 4 shows this clearly. The failure reflects an implicit prior over garment shape that is incompatible with silhouettes whose intended appearance depends on free-hanging volume and off-body projection, both properties absent from Western training data.

Tonal drift is observed in outputs for Monk Scale 7 subjects as subtle skin-tone shifts at garment-body boundaries, where dark skin meets high-contrast textile patterns. The artefact likely reflects inconsistencies in the compositing of the synthesised garment region with the preserved person region during inpainting-based try-on. Systematic quantitative evaluation across the full Monk 6–10 range is deferred to future work, but the observation motivates skin-tone-stratified evaluation as a component of future AfriVTON-Bench releases.

4.5 Discussion

The results converge on a single root cause: the Western centricity of the training data underlying current VTON systems. That the degradation is architecture-agnostic – replicated across a DWPose-conditioned diffusion model and an

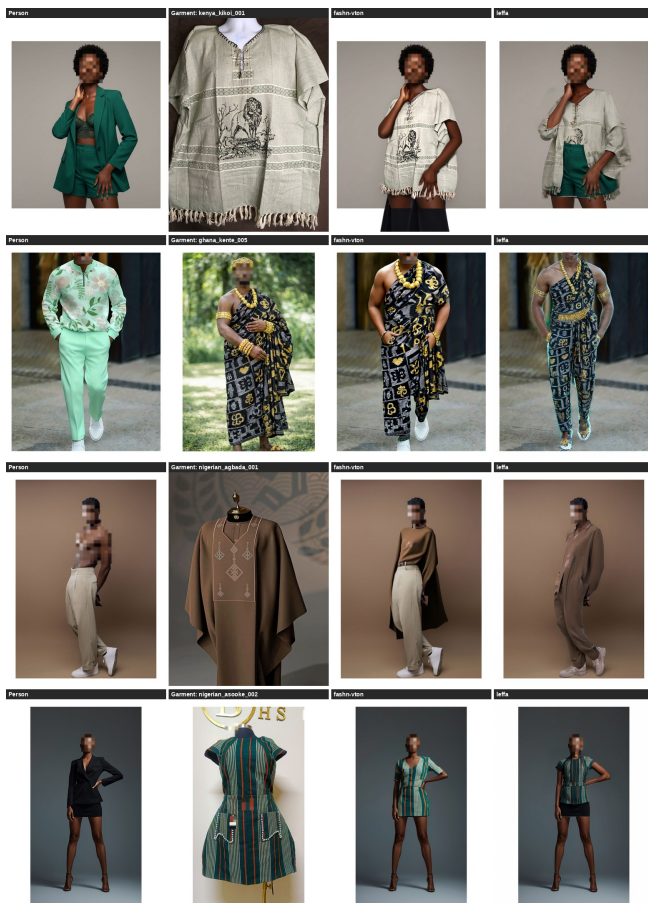


Figure 4: Qualitative comparison across four garment categories. Each row shows (left to right) the input person, the reference garment, the FASHN-VTON output, and the LEFFA output. Rows from top: Kenya Kikoi (draped, intermediate complexity), Ghana Kente (high-frequency woven pattern, worst-performing category), Nigeria Agbada (voluminous silhouette), Nigeria Aso-Oke (geometric woven stripe). Faces blurred for privacy.

attention-flow warping model – rules out model-level explanations and directly implicates the data distribution. The failure also scales with garment visual complexity rather than with geographic origin, which has a useful implication for the field: African garments are not a niche edge case but a principled stress test that surfaces generalisation failures of broader relevance.

As global e-commerce platforms expand into African markets, VTON systems that perform poorly on locally worn garments risk providing a substantially degraded shopping experience for African consumers. More broadly, AfriVTON-Bench demonstrates that benchmark diversity is a prerequisite for trustworthy capability assessment in generative vision, and that systems which appear state-of-the-art on existing benchmarks may fail substantially on garment types worn by the majority of the world’s population.



Figure 5: Ghana Kente try-on output illustrating pattern dissolution. Left to right: input person, reference Kente garment, FASHN-VTON output, LEFFA output. The characteristic Adinkra symbol motifs are substantially distorted in both outputs; LEFFA additionally collapses the draped silhouette into fitted trousers.

5 Conclusion

AfriVTON-Bench is the first benchmark designed to evaluate VTON robustness on African garment types and skin tone diversity. Evaluating FASHN-VTON and LEFFA across 1,012 person-garment pairs reveals systematic, architecture-agnostic performance degradation: a 9–10 point gSSIM deficit on African garments relative to Western controls, with Kente and Ankara exhibiting deficits of 17–19 points. Three failure modes are characterised – pattern dissolution, drape collapse, and tonal drift – all attributable to the Western-centricity of existing VTON training corpora. The dataset, evaluation code, and model outputs are released publicly to support reproducible research on culturally inclusive garment synthesis.

Ethical Considerations

This work involves the collection, processing, and public release of person images sourced from the web. All person images were processed through a multi-detector face blurring pipeline before use in experiments or publication. No face is visible in any released image. Garment images were sourced from publicly available web resources and Wikimedia Commons; no personally identifiable information is associated with garment-only images.

The construction of AfriVTON-Bench was motivated by the recognised risk that AI systems trained on non-representative data reproduce inequities in the populations they serve. We acknowledge that our person image set is small and skewed toward Monk 7, limiting the conclusions that can currently be drawn from tonal drift observations; future versions will include a broader demographic range. The models evaluated are publicly released systems and our evaluation is adversarial in intent: we test failure modes rather than deploy the systems in consumer-facing applications. The failure modes reported, particularly tonal drift, should be interpreted as signals for future work rather than as grounds for current deployment decisions.

AI-assisted writing tools were used during the preparation of this manuscript. All content is the original work of the authors, and the authors take full responsibility for the accuracy and integrity of the submission.

Limitations

The primary limitation of AfriVTON-Bench in its current form is the size of the person image set. Fifteen images, predominantly at Monk Scale 7, provide insufficient statistical power for robust skin-tone-stratified analysis. Evaluation is also restricted to two architectures; the assessment does not yet include the diffusion transformer models that currently lead standard benchmarks. The benchmark lacks user study data, meaning synthesis quality is assessed entirely through reference-based metrics without perceptual ground truth from human evaluators.

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